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$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x(1 - \cos x)} \left(= \frac{0}{0} \right) \\ & \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos x - (\cos x - x \sin x)}{(1 - \cos x) + (x \sin x)} = \lim_{x \rightarrow 0} \frac{x \sin x}{(1 - \cos x) + (x \sin x)} \left(= \frac{0}{0} \right) \\ & \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\sin x + x \cos x}{\sin x + x \cos x + \sin x} = \lim_{x \rightarrow 0} \frac{\sin x + x \cos x}{2 \sin x + x \cos x} \left(= \frac{0}{0} \right) \\ & \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos x + (\cos x - x \sin x)}{2 \cos x + (\cos x - x \sin x)} = \lim_{x \rightarrow 0} \frac{2 \cos x - x \sin x}{3 \cos x - x \sin x} = \frac{2}{3} \end{aligned}$$

References